Vagus Nerve Stimulation

Anatomy, Mechanisms, Stimulation Parameters, Biomarkers, and Clinical Translation

Abstract

Vagus nerve stimulation (VNS) has evolved from an implanted therapy for drug-resistant epilepsy into a broad family of bioelectronic interventions that target vagal pathways at cervical, auricular, and abdominal sites. The rapid expansion of non-invasive approaches—particularly transcutaneous auricular VNS (taVNS) and transcutaneous cervical VNS (tcVNS)—has created substantial interest in their capacity to modulate brainstem neuromodulatory systems, interoceptive processing, autonomic control, neuroimmune signaling, and large-scale brain networks. At the same time, the field remains methodologically heterogeneous: electrode location, waveform, pulse width, frequency, current intensity, duty cycle, laterality, session duration, sham condition, respiratory state, and outcome measures differ markedly across studies. This focused narrative review synthesizes anatomical, physiological, neuroimaging, psychophysiological, and clinical evidence with particular emphasis on non-invasive VNS. Evidence is strongest for anatomical access to central vagal projections through the auricular branch, reproducible engagement of the nucleus tractus solitarius and related brainstem networks in selected imaging paradigms, and a generally favorable short-term safety profile. However, commonly used surrogate outcomes are not equivalent: acute vagally mediated heart-rate variability has not emerged as a reliable universal target-engagement biomarker, while pupil responses appear protocol-dependent and are more consistently observed with brief phasic stimulation. Clinical evidence is promising but uneven across depression, epilepsy, insomnia, pain, post-stroke rehabilitation, disorders of consciousness, and inflammatory conditions. The major translational challenge is therefore not simply whether VNS can produce biological effects, but how to establish target engagement, dose, state dependence, responder profiles, and clinically meaningful reproducibility.

Keywords: vagus nerve stimulation; taVNS; tcVNS; interoception; nucleus tractus solitarius; locus coeruleus; neuromodulation; bioelectronic medicine; biomarkers

1. Introduction

The vagus nerve (cranial nerve X) is the principal parasympathetic conduit linking the brainstem with thoracic and abdominal organs. It is a mixed nerve with a strong sensory bias: approximately 80% of fibers are afferent and 20% are efferent, enabling continuous bidirectional

communication between internal organs and the central nervous system [1,2]. Vagal afferents convey mechanical, chemical, metabolic, inflammatory, and visceral state information to the medulla, while efferent pathways contribute to cardiac, respiratory, gastrointestinal, endocrine, and immune regulation [1–3].

This anatomy makes the vagus an unusual neuromodulation target. Electrical stimulation can influence local peripheral physiology while also engaging central circuits through the nucleus tractus solitarius (NTS), parabrachial complex, locus coeruleus (LC), dorsal raphe, hypothalamus, thalamus, amygdala, insula, anterior cingulate cortex, and related networks [1,7–11]. The same broad reach that makes VNS attractive also makes mechanistic interpretation difficult: changes in symptoms or physiology can arise through overlapping afferent, efferent, autonomic, neuromodulatory, and neuroimmune pathways.

The modern VNS literature therefore needs to distinguish three questions: whether the intended nerve fibers are engaged, which neural and physiological systems are subsequently modulated, and whether those changes translate into durable clinical benefit. This distinction is especially important for non-invasive stimulation, where the electrical field also interacts with non-vagal cutaneous nerves and where sensory experience can compromise blinding [4–7].

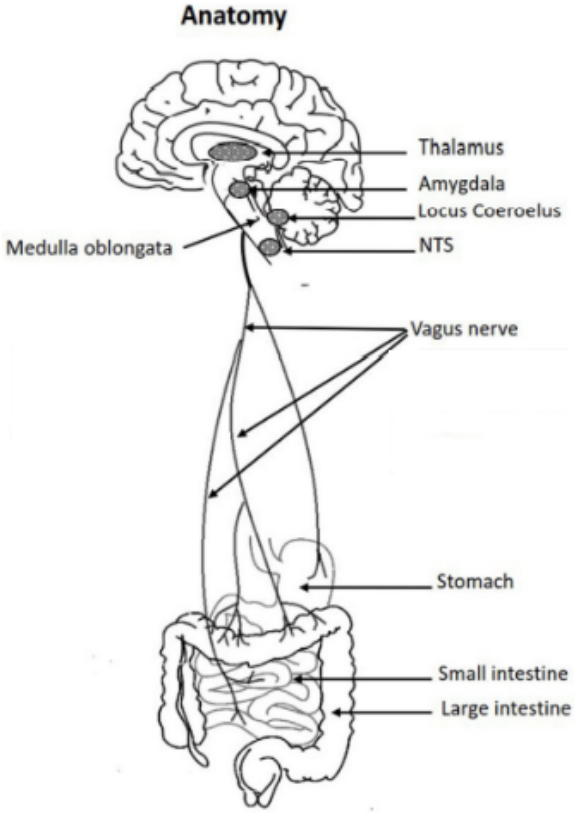


Figure 1. Basic vagal anatomy and central projections. The vagus conveys visceral information to the NTS and broader brainstem/forebrain networks while providing descending autonomic output to visceral organs. Reproduced from Breit et al. (2018) [1], Frontiers in Psychiatry, CC BY 4.0.

2. Anatomical and physiological basis

2.1 Afferent and efferent organization

The major visceral sensory cell bodies of the vagus are located in the nodose (inferior) ganglion, whereas jugular (superior) ganglion fibers contribute more somatic sensory input. Visceral afferents project predominantly to the NTS, which serves as the first major central relay and integrates vagal input with respiratory, cardiovascular, metabolic, and inflammatory information. Efferent parasympathetic fibers arise mainly from the dorsal motor nucleus of the vagus and nucleus ambiguus [2,4].

At the organ level, vagal signaling is functionally diverse. The nerve contributes to satiety and nutrient sensing, gastrointestinal motility and secretion, cardiopulmonary reflexes, and immune regulation. Importantly, the simplified “parasympathetic output nerve” description is incomplete: the dominant afferent component means that many effects of VNS are initiated by ascending sensory traffic rather than direct efferent stimulation [1–3].

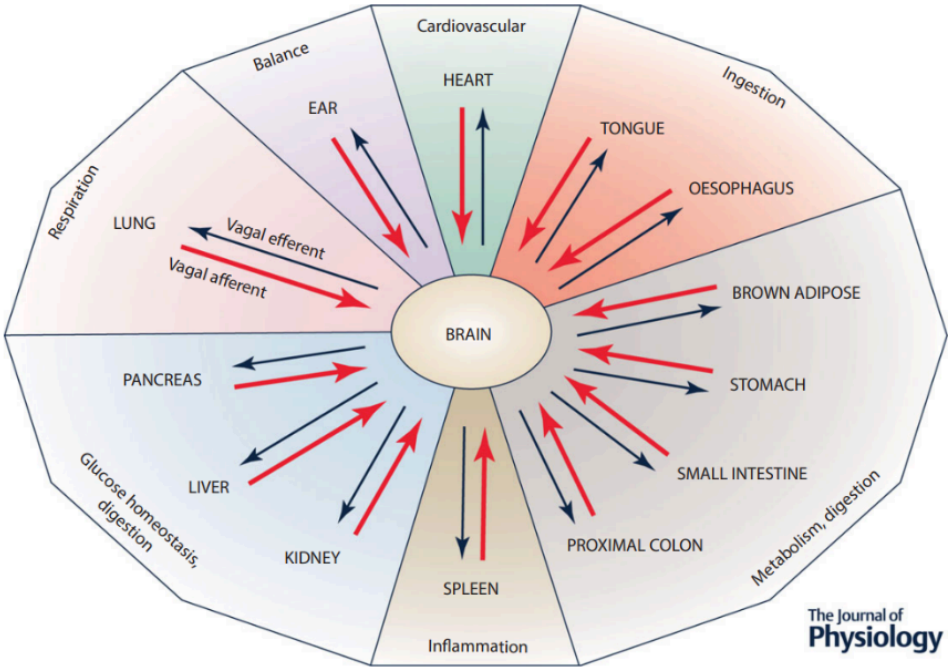


Figure 2. Bidirectional vagal communication between brain and peripheral organs. Red arrows indicate afferent traffic and dark arrows efferent traffic. Source: de Lartigue (2016) [3], The Journal of Physiology.

2.2 The auricular branch and why the ear is a target

The auricular branch of the vagus nerve (ABVN, Arnold’s nerve) provides a superficial route to vagal afferents in the external ear. The classic cadaveric map by Peuker and Filler was based on only 14 ears from seven cadavers and demonstrated heterogeneous innervation with substantial overlap among the vagus, great auricular, auriculotemporal, and lesser occipital nerves [5]. Subsequent anatomical reviews therefore caution against treating any visible auricular territory as a perfectly selective “vagus-only” zone [4].

The cymba conchae has nevertheless become a common active target because anatomical and fMRI work supports vagal access from this region [4,8,9]. The tragus and cavum conchae are also used, but stimulation location should be interpreted together with electrode geometry,

current path, intensity, and sham design. The practical implication is that electrode placement is not a trivial methodological detail—it is part of the experimental dose.

3. VNS modalities: invasive and non-invasive routes

Modality	Primary target / access	Typical strengths	Key limitations
Implanted cervical VNS (iVNS)	Surgically placed helical electrodes around the cervical vagus; pulse generator in chest	Known nerve contact; programmable long-term therapy; established evidence in epilepsy and selected other indications	Surgery, cost, hardware-related adverse events; fiber recruitment is not perfectly selective
Transcutaneous cervical VNS (tcVNS)	Electrical field delivered through skin at the neck over the cervical vagus	Portable, non-invasive; clinically developed for headache disorders	Current traverses skin/muscle; differs biophysically from implanted VNS; should not be equated with taVNS
Transcutaneous auricular VNS (taVNS)	Surface electrodes at cymba conchae, tragus or concha targeting ABVN	Non-invasive, scalable, compatible with EEG/fMRI and repeated sessions	Auricular innervation overlap; uncertain dose at nerve; sensory confounds; no universal stimulation standard
Percutaneous auricular approaches	Needle/electrode systems placed near auricular nerves	Longer-duration peripheral stimulation without cervical surgery	Different intervention class; device- and indication-specific evidence

These routes should not be pooled casually in evidence synthesis. tcVNS stimulates the cervical bundle through the neck, whereas taVNS targets the ABVN at the ear. They differ in anatomy, waveform, current density, recruited fibers, sensory experience, device engineering, and regulatory history [6,30]. In the United States, cervical non-invasive devices have regulatory clearances for headache indications, whereas auricular taVNS remains investigational for most neurological and psychiatric applications [6,30].

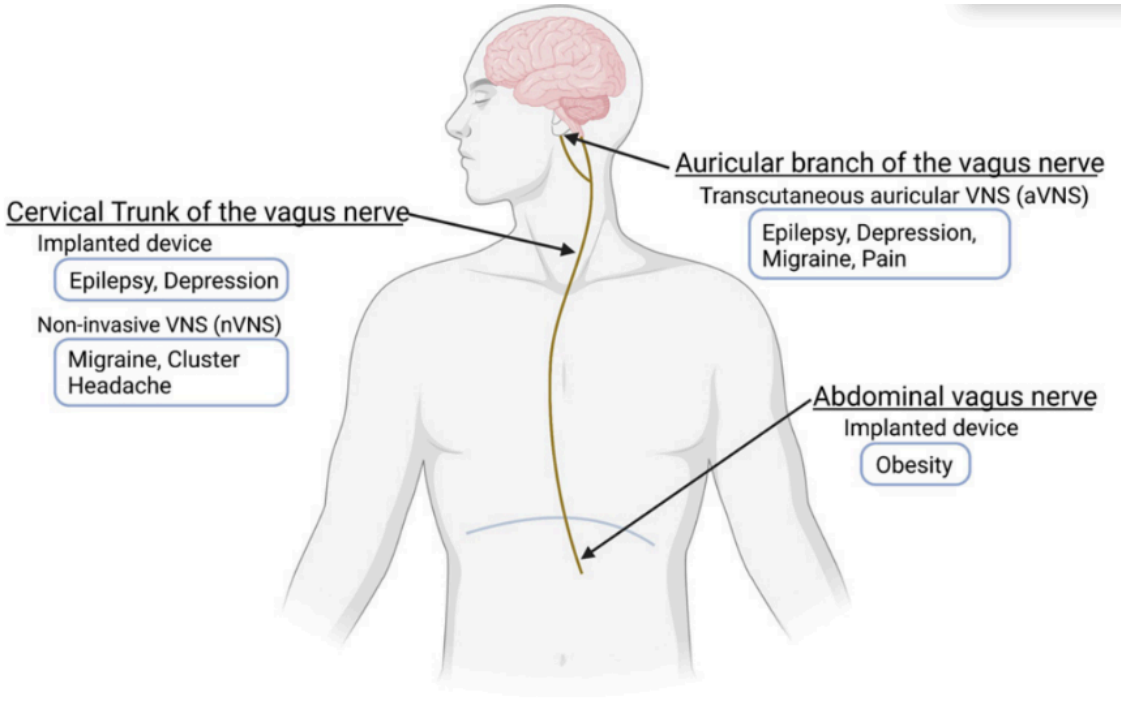


Figure 3. Overview of VNS access routes and selected clinical uses as summarized by Goggins et al. (2022) [2], CC BY 4.0. Important: the figure is a 2022 literature snapshot rather than a current regulatory map; indications and device authorizations are jurisdiction- and device-specific and have continued to evolve.

4. Stimulation dose is multidimensional

A recurring weakness in the tVNS literature is the tendency to describe “dose” using frequency alone. In reality, biological exposure depends on a multidimensional set of variables: anatomical site, electrode contact area and orientation, waveform, pulse width, frequency, current amplitude, charge per pulse, train duration, duty cycle, session length, laterality, skin–electrode impedance, and the participant’s physiological state [6,7]. The international minimum reporting recommendations were created precisely because incomplete reporting has made replication and cross-study comparison difficult [6].

Variable	Common choices in human taVNS research	Why it matters
Target site	Cymba conchae, tragus, cavum/concha; earlobe or superior scapha often used as control	Auricular innervation is heterogeneous; location changes likely neural recruitment
Frequency	Often ~20–30 Hz; lower and higher frequencies also used	Changes temporal recruitment and perceived sensation; no single optimal frequency established
Pulse width	Frequently ~200–500 μs	Affects charge per pulse and recruitment threshold

Intensity	Fixed mA, perceptual threshold multiples, or “strong but non-painful” individualized dosing	Absolute mA is not comparable across electrodes/skin impedance; sensation can unblind conditions
Pattern	Continuous, intermittent, short phasic bursts, respiratory-gated trains	Pattern can change neuromodulatory and biomarker effects; phasic protocols appear particularly relevant for pupil responses
Laterality	Left, right, or bilateral	Historical preference for left-sided cervical VNS does not automatically transfer to auricular stimulation
Physiological state	Rest, task, sleep, respiration phase	State can gate NTS and network responsiveness; exhalation is one demonstrated example [11]
Sham	Earlobe, superior scapha, alternative nerve territory, inactive electrodes	No sham is biologically perfect; matched sensation and credible blinding are essential

INTERPRETATION RULE
A stimulation protocol should be treated as a complete intervention specification—not as “25 Hz taVNS.” Two studies using the same frequency may deliver very different effective doses because site, pulse width, intensity, electrode geometry, pattern, and physiological context differ.

5. Evidence for central target engagement

5.1 Brainstem and fMRI evidence

Human neuroimaging provides some of the strongest evidence that auricular stimulation can access classical central vagal projections. Frangos et al. reported that left cymba conchae stimulation, relative to earlobe control, engaged the ipsilateral NTS and other regions compatible with central vagal pathways [8]. Yakunina et al. compared several auricular locations and found that cymba conchae stimulation produced particularly strong NTS and LC responses relative to sham [9]. Badran et al. extended concurrent taVNS/fMRI work using tragus stimulation, while emphasizing that differences across earlier studies likely reflected variation in targets and parameters [10].

Brainstem imaging remains technically difficult because the NTS and LC are small structures located near pulsatile vessels and cerebrospinal fluid spaces. This creates susceptibility to physiological noise, registration error, and analytical choices. Brainstem activation should therefore be considered informative but not a universally validated “on/off” biomarker of successful vagal engagement.

5.2 Respiratory state can gate stimulation effects

Sclocco et al. used 7-T fMRI and respiration-gated auricular vagal afferent nerve stimulation to test whether the phase of breathing changes brainstem response. Exhalation-gated stimulation produced stronger responses in the NTS region and monoaminergic nuclei than inhalation-gated stimulation and also altered cardiovagal output [11]. This study is important conceptually because it demonstrates that the same electrical parameters can produce different effects depending on the state of the organism at the moment stimulation is delivered.

5.3 Cortical and large-scale network effects

From the NTS, ascending pathways can recruit LC and raphe systems and influence thalamic, limbic, insular, cingulate, prefrontal, hippocampal, and striatal regions. Network-level effects are therefore plausible downstream consequences of vagal afferent stimulation rather than evidence that current directly reaches cortex [8–12]. In depression, Fang et al. observed increased left anterior insula response during active stimulation, and early insular activation was associated with subsequent symptom improvement after four weeks [12]. This type of result is clinically interesting because it links an acute neural response to longitudinal outcome, but replication and prospective validation are still required.

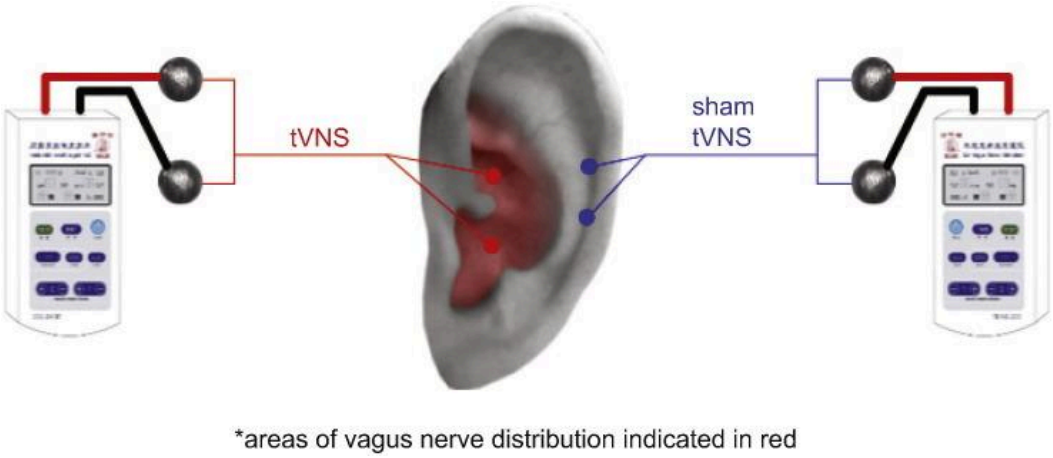


Figure 4. Active and sham auricular stimulation locations used in a longitudinal depression neuroimaging study. Reproduced from Fang et al. (2017) [12], NeuroImage: Clinical, open-access CC BY-NC-ND 4.0.

6. Interoception, body–brain coupling, and allostatic relevance

Because most vagal fibers are afferent, taVNS is particularly relevant to interoception—the sensing, representation, and interpretation of internal bodily signals. In a within-subject experiment, Villani et al. found that active taVNS improved performance on a heartbeat discrimination task and increased confidence, while not improving heartbeat counting performance or metacognitive awareness [13]. The selective pattern matters: it suggests modulation of particular interoceptive computations rather than a generic increase in “body awareness.”

Gastrointestinal signals provide another window on body–brain communication. Müller et al. used electrogastrography and neuroimaging in a randomized crossover design and found that

taVNS increased stomach–brain coupling in the NTS/midbrain and broader cortical network; cortical changes were related to changes in hunger ratings [14]. Together with cardiac interoception findings, these studies support the use of taVNS as an experimental perturbation of ascending visceral signaling. They do not, however, imply that taVNS reproduces all endogenous interoceptive signals or that every observed subjective change is vagus-specific.

7. Candidate biomarkers: what is convincing and what is not?

Candidate marker	Current interpretation	Strength / caveat
Brainstem fMRI (NTS/LC)	Directly aligned with proposed afferent pathway; positive evidence in selected studies	High mechanistic value but technically difficult; small nuclei and physiological noise
Pupil dilation	Candidate marker of LC–noradrenergic engagement	Protocol-dependent. 2025 living Bayesian meta-analysis found strong evidence for dilation with brief phasic stimulation, but not conventional long stimulation [15–18]
EEG / alpha dynamics	Acute taVNS can attenuate alpha oscillations in some paradigms	Potentially useful but not specific to vagal engagement and sensitive to arousal/sensory effects [15]
vmHRV / RMSSD / HF-HRV	Often treated as cardiac vagal readout	Not a reliable universal acute biomarker: a living Bayesian meta-analysis found strong evidence for no acute active-vs-sham effect in healthy samples [19]
Interoceptive behavior	Heartbeat discrimination, confidence, visceral perception	Mechanistically relevant but task validity and specificity need care [13]
Gastric coupling / EGG-fMRI	Directly linked to vagal gut–brain signaling	Promising mechanistic measure, currently specialized and not yet a routine clinical biomarker [14]
Inflammatory markers	Cytokines, immune cell signaling	Biologically plausible via inflammatory reflex pathways; clinical specificity and optimal timing remain unresolved [31,32]

CRITICAL EVIDENCE POINT

Heart-rate variability should not be used as a universal positive control for acute taVNS target engagement. The strongest synthesis to date found evidence for the null effect on vagally mediated HRV in healthy participants [19].

8. Neurochemical and neuroimmune mechanisms

8.1 NTS–LC and NTS–raphe pathways

One influential mechanistic model proposes that vagal afferent activity reaches the NTS and then recruits neuromodulatory nuclei, including the noradrenergic LC and serotonergic raphe complex. These nuclei have diffuse projections and can alter arousal, attention, learning, sensory gain, plasticity, and affective processing. Human fMRI, pupillometry, and EEG provide convergent but not perfectly consistent evidence for this pathway [8–11,15–18]. Importantly, pupil dilation is not a pure LC assay; sensory salience, luminance, effort, and arousal can also change pupil size.

8.2 The inflammatory reflex

Vagal signaling also participates in neuroimmune control. Preclinical work established the concept of an inflammatory reflex in which neural sensing of inflammatory signals is linked to efferent pathways capable of suppressing excessive cytokine production [31,32]. The splenic pathway is indirect and involves interactions among vagal, sympathetic, and cholinergic immune mechanisms rather than a simple direct vagal innervation of all immune targets. This distinction is important when translating anti-inflammatory effects from implanted VNS or animal models to auricular stimulation in humans.

9. Clinical evidence: where the field stands

The clinical literature is best interpreted by indication and by stimulation route. Evidence from implanted VNS, tcVNS, and taVNS should not be merged simply because all are labeled “VNS.” The table below emphasizes representative human taVNS evidence and explicitly flags when conclusions remain preliminary.

Indication	Representative evidence	What can reasonably be concluded
Major depressive disorder	Pilot and neuroimaging studies [12,21]; 2023 RCT meta-analysis included 12 studies / 838 participants [22]	Depression scores generally favor taVNS, but evidence quality in the meta-analysis was low to very low; larger rigorous trials are needed
Epilepsy	2024 meta-analysis of clinical taVNS/auricular interventions [23]	Potential reduction in seizure burden, but included interventions and study quality are heterogeneous; not equivalent to established implanted VNS evidence
Chronic insomnia / sleep disorders	2024 randomized clinical trial [24]; 2026 systematic review/meta-analysis [25]	Promising patient-reported sleep benefits with repeated treatment; heterogeneity and protocol standardization remain important
Disorders of consciousness	Small pilot work including perfusion imaging [26]; later reviews emphasize limited controlled evidence	Biologically plausible but still exploratory; small samples and subgroup effects preclude routine clinical conclusions

Pain / fibromyalgia	Protocol-level and smaller clinical studies; growing 2026 review literature [27,28]	Potential analgesic and autonomic/immune effects, but condition-specific high-quality RCT evidence is limited
Interoceptive / metabolic modulation	Heartbeat discrimination [13]; stomach–brain coupling [14]	Strongest value currently mechanistic rather than therapeutic; demonstrates modifiable body–brain signaling
Post-stroke rehabilitation	Growing non-invasive literature; strongest regulatory evidence in stroke is for implanted paired VNS, not taVNS	Do not extrapolate implanted paired-VNS efficacy directly to auricular stimulation
Migraine / cluster headache	Substantial VNS literature, but clinically authorized non-invasive evidence is primarily cervical tcVNS	Route must be specified; tcVNS efficacy does not establish taVNS efficacy

9.1 Depression

Depression is among the most studied psychiatric indications for taVNS. Early pilot work by Rong et al. reported improvement during repeated auricular stimulation [21], and Fang et al. identified anterior insula activity as a candidate predictor of longitudinal response [12]. A 2023 systematic review and meta-analysis of randomized controlled trials included 12 studies with 838 participants and found reductions in depression severity, while grading important efficacy outcomes as low or very low certainty [22]. The appropriate conclusion is therefore “promising but not definitive,” rather than equivalence to established antidepressant treatments.

9.2 Epilepsy

Implanted cervical VNS has a mature evidence base in drug-resistant epilepsy, but auricular taVNS is a separate intervention. A 2024 meta-analysis reported a higher response rate with taVNS/auricular interventions than control, yet the included literature was heterogeneous and incorporated forms of auricular stimulation with variable correspondence to modern electrical taVNS [23]. This limits direct inference about a standardized taVNS protocol.

9.3 Insomnia and sleep

Sleep disorders have become an active taVNS area. A 2024 randomized clinical trial in chronic insomnia used 30-minute sessions twice daily, five days per week for eight weeks and reported superior improvements in patient-reported outcomes during treatment [24]. A 2026 systematic review and meta-analysis concluded that taVNS may improve sleep quality as well as anxiety/depression outcomes, while emphasizing between-study heterogeneity [25]. These results justify further confirmatory trials with objective sleep measures and standardized stimulation reporting.

9.4 Pain and fibromyalgia

Vagal modulation of nociception can involve brainstem descending pain-control systems, autonomic regulation, arousal, and neuroimmune pathways. In fibromyalgia, a published

protocol compared auricular and cervical non-invasive stimulation with sham/control montages and included pain, fatigue, quality of life, sleep, depression, HRV, and inflammatory measures [27]. The design is useful methodologically because it illustrates how route-specific and non-vagal control conditions can separate generalized electrical stimulation effects from vagal mechanisms.

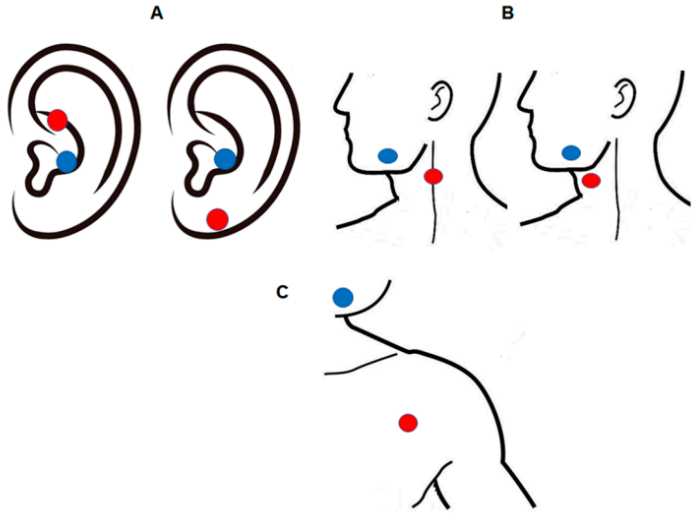


Figure 5. Example electrode montages comparing auricular, cervical, sham, and non-vagal control stimulation in a fibromyalgia protocol. Reproduced from Molero-Chamizo et al. (2022) [27], Brain Sciences, CC BY 4.0.

9.5 Disorders of consciousness

Pilot work in disorders of consciousness has reported cerebral blood-flow changes and possible behavioral responses after repeated taVNS [26]. More recent syntheses remain cautious: sample sizes are small, diagnostic groups are heterogeneous, stimulation protocols vary, and overall between-group efficacy has not been established robustly. taVNS in this population should therefore be treated as investigational and mechanistically interesting, not as a validated consciousness-restoration therapy.

10. Safety and tolerability

Non-invasive auricular stimulation is generally well tolerated in research settings. A systematic review and meta-analysis evaluated 177 studies involving 6,322 participants and found no clear excess of adverse-event risk with active taVNS compared with controls; commonly reported events included local ear discomfort, headache, tingling, and skin irritation [20]. However, the same review exposed a major reporting problem: more than half of included studies did not explicitly report whether adverse events occurred. Apparent safety should therefore not be confused with complete safety surveillance.

Record adverse events systematically rather than only reporting spontaneous complaints.

Document stimulation site, skin condition, intensity calibration, session duration, and discontinuations.

Screen for clinically relevant cardiac, neurological, dermatological, and device-specific contraindications according to the protocol and local ethics requirements.

Report severe events even when investigators judge them unrelated to stimulation.

11. Reproducibility problems and sources of heterogeneity

Problem	Why it distorts the literature	Better practice
Inconsistent terminology	“tVNS” can mean auricular or cervical stimulation	Always specify taVNS vs tcVNS and anatomical site
Incomplete parameter reporting	Prevents dose reconstruction and replication	Follow minimum reporting standards [6]
Variable sham conditions	Sensation and non-vagal nerve activation differ across controls	Assess sham credibility, sensation ratings, and blinding
Assuming earlobe is inert	Earlobe is non-vagal relative to common maps but not physiologically “inactive”	Interpret as active control territory, not biological zero
Small brainstem nuclei	fMRI is vulnerable to physiological noise and registration error	Use brainstem-optimized acquisition/analysis and report ROIs transparently
Over-reliance on HRV	Acute HRV effects are not robust across controlled studies	Use multimodal target-engagement evidence rather than one surrogate [19]
Disease and protocol heterogeneity	Prevents meaningful pooling across studies	Predefine clinically coherent populations and dose parameters
Sensory confounding	Higher intensity may cause stronger sensation and alter arousal/pupil/behavior	Percept-match conditions where feasible and model sensation explicitly

12. Current directions

The field is moving away from the question “does taVNS work?” toward more precise questions about when, where, how, and in whom stimulation changes a defined biological process. Three directions are particularly important. First, multimodal target engagement should combine measures that sit at different levels of the proposed pathway—for example, brainstem fMRI or EEG with pupillometry, respiration, ECG, and behavioral readouts—rather than relying on a single surrogate. Second, stimulation should be treated as state-dependent: respiratory phase, arousal, sleep/wake state, task context, and baseline autonomic function may change responsiveness [11,18]. Third, dose optimization should be based on mechanistic response surfaces rather than inherited “standard settings” that differ across devices and historical studies [6,7,29].

Recent reviews published in 2026 emphasize the same translational gap: the literature is expanding faster than protocol standardization, and biomarker-guided personalization is likely to be more informative than simply increasing the number of indications tested [28,29]. This is a healthy shift because it places causal mechanism and reproducibility before broad therapeutic claims.

13. Conclusion

Vagus nerve stimulation is best understood as a family of interventions acting on a complex bidirectional body–brain system. The evidence strongly supports the vagus as a biologically powerful target and supports non-invasive access to central vagal pathways through the external ear in humans. Yet the central challenge for taVNS is selectivity and reproducibility: anatomical overlap, heterogeneous electrical dose, state dependence, imperfect sham conditions, and uncertain biomarkers make apparently similar studies difficult to compare. The most credible path forward is therefore mechanism-first—verify target engagement, quantify dose and physiological context, use appropriate controls, and then test whether the identified biological response predicts durable clinical benefit. Within that framework, taVNS is not merely a lower-risk substitute for implanted VNS; it is a distinct neuromodulation modality whose advantages will depend on how precisely its afferent pathways can be engaged and measured.

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